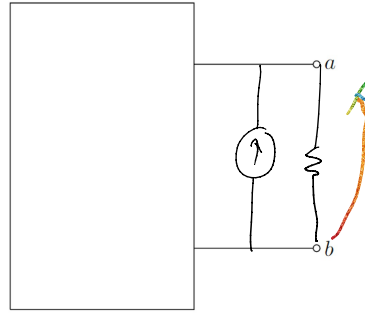


L16 - Norton's Theorem

Thursday, March 14, 2024 11:09 AM

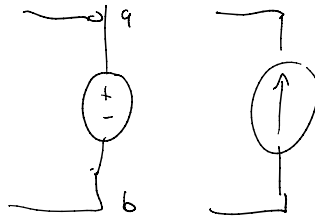
Core Principles

- I_N is current between the short across terminals
 - Flows from b to a
- R_N is equivalent resistance across terminals
 - Turn off all independent sources
- If there is a dependent voltage source, must add a test source (1V or 1A)
 - $R_N = V_{test} / I_{test}$



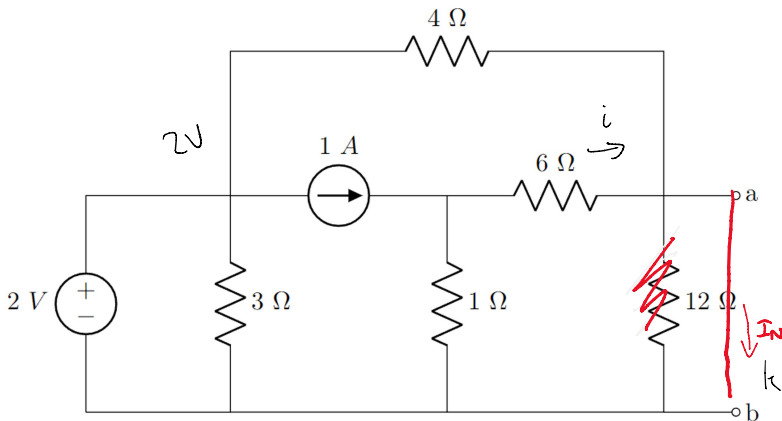
$$R_{TH} = R_N$$

methods identical



Steps:

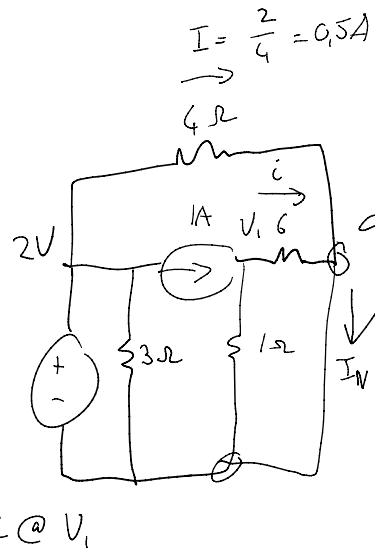
- Short between terminals a-b
- Find I_N , the current flowing from b to a
- Turn off all independent sources
- Is there a dependent source?
 - If no, find equivalent resistance for R_N
 - If yes, draw a 1V or 1A source between terminals
 - Calculate the current through or voltage across that source
 - Use ohms law to find R_N



$$\begin{aligned} \text{KCL @ } i &= \frac{6}{\frac{7}{6}} \\ &= \frac{1}{7} \text{ A} \end{aligned}$$

$$-0,5 - \frac{1}{7} + I_N = 0$$

$$\therefore I_N = 0,5 + \frac{1}{7} = 0,64 \text{ A}$$

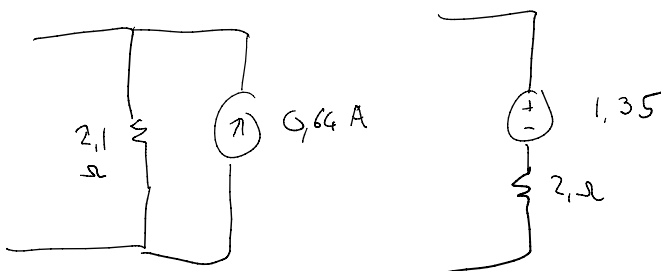


$$\frac{V_1}{1} + \frac{V_1}{6} - 1 = 0$$

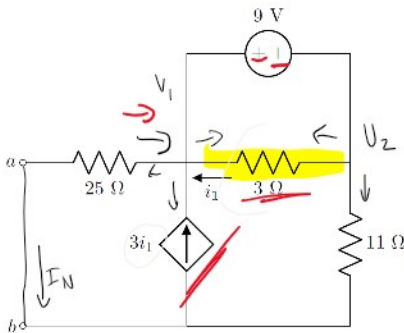
$$6V_1 + V_1 - 6 = 0$$

$$7V_1 = 6$$

$$V_1 = \frac{6}{7}$$



$$\frac{V_{TH}}{R_{TH}} = \frac{1,35}{2,1} = 0,64 = I_N$$



$$\textcircled{1} \quad V_1 - V_2 = 9V$$

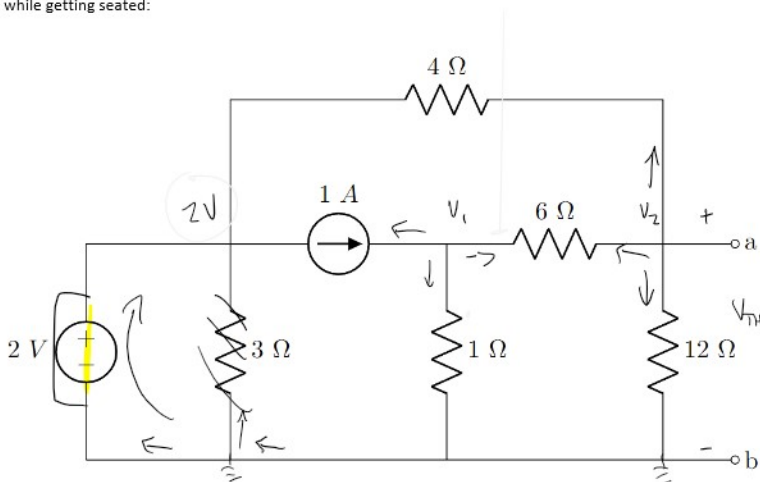
$$\frac{V_2}{11} + \frac{V_1}{25} - 3i_1 + \frac{V_1 - V_2}{3} + \frac{V_2 - V_1}{3} = 0$$

$$i_1 = \frac{V_2 - V_1}{3}$$

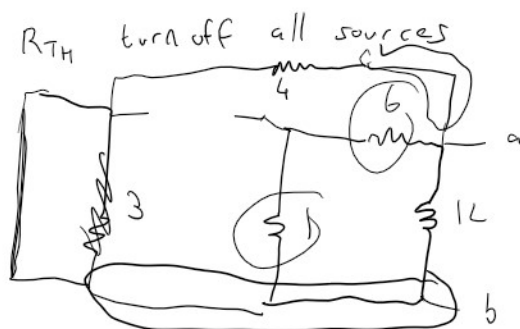
$$\frac{V_2}{11} + \frac{V_1}{25} - 3\left(\frac{V_2 - V_1}{3}\right) = 0$$

$$I_N = \frac{V_1}{25}$$

Do while getting seated:



Find V_{TH} and R_{TH} across terminals a-b



$$\textcircled{1} \quad V_{TH} = V_2$$

for KCL @ V_1

$$-1 + \frac{V_1}{1} + \frac{V_1 - V_2}{6} = 0$$

$$-6 + 6V_1 + V_1 - V_2 = 0$$

$$7V_1 - V_2 = 6$$

for KCL @ V_2

$$\frac{V_2 - 2}{4} + \frac{V_2 - V_1}{6} + \frac{V_2}{12} = 0$$

$$3V_2 - 6 + 2V_2 - 2V_1 + V_2 = 0$$

$$-2V_1 + 6V_2 = 6$$

$$V_{TH} = 1,35V$$

$$(6+1) \parallel (4) \parallel 12 = R_{TH}$$

$$= 2,1 \Omega$$